

2026 年计算机类转专业（含院内专业调整、图灵班、智班选拔，物理计算机交叉）上机测试

比赛信息与统计

开始时间	2026-05-15 13:00:00
结束时间	2026-05-15 15:00:00

题目 ID	标题	通过率	通过人数	尝试人数
A	制作蛋糕	98%	137	140
B	Hardwood Species	79%	93	118
C	Safecracker	85%	68	80
D	购买优惠券	62%	69	111
E	合并编码	70%	64	92
F	网线主管	60%	58	96
G	Dungeon Master	84%	58	69
H	糖果	72%	49	68

1 A: 制作蛋糕

总时间限制: 1000ms; 内存限制: 65536kB

1.1 描述

小 A 擅长制作香蕉蛋糕和巧克力蛋糕。制作一个香蕉蛋糕需要 2 个单位的香蕉, 250 个单位的面粉, 75 个单位的糖, 100 个单位的黄油。制作一个巧克力蛋糕需要 75 个单位的可可粉, 200 个单位的面粉, 150 个单位的糖, 150 个单位的黄油。一个香蕉蛋糕可以卖出 400 元, 而一个巧克力蛋糕可以卖出 450 元。为了避免蛋糕变质, 每种蛋糕至多只能制作 100 个。

现已知每种原料的数量, 求小 A 至多可以卖出多少元的蛋糕。

1.2 输入

依次输入面粉、香蕉、糖、黄油、可可粉的数量, 每种原料数量均为不超过 10^5 的整数。

1.3 输出

依次输出卖出的钱数、香蕉蛋糕数量、巧克力蛋糕数量, 每个数占一行。

1.4 样例输入

```
4000
6
2000
500
500
```

1.5 样例输出

```
1700
2
2
```

2 B: Hardwood Species

总时间限制: 10000ms; 内存限制: 65536kB

2.1 描述

Hardwoods are the botanical group of trees that have broad leaves, produce a fruit or nut, and generally go dormant in the winter. America's temperate climates produce forests with hundreds of hardwood species – trees that share certain biological characteristics. Although oak, maple and cherry all are types of hardwood trees, for example, they are different species. Together, all the hardwood species represent 40 percent of the trees in the United States.

On the other hand, softwoods, or conifers, from the Latin word meaning "cone-bearing," have needles. Widely available US softwoods include cedar, fir, hemlock, pine, redwood, spruce and cypress. In a home, the softwoods are used primarily as structural lumber such as 2x4s and 2x6s, with some limited decorative applications.

Using satellite imaging technology, the Department of Natural Resources has compiled an inventory of every tree standing on a particular day. You are to compute the total fraction of the tree population represented by each species.

2.2 输入

Input to your program consists of a list of the species of every tree observed by the satellite; one tree per line. No species name exceeds 30 characters. There are no more than 10,000 species and no more than 1,000,000 trees.

2.3 输出

Print the name of each species represented in the population, in alphabetical order, followed by the percentage of the population it represents, to 4 decimal places.

2.4 样例输入

```
Red Alder
Ash
Aspen
Basswood
Ash
Beech
Yellow Birch
Ash
Cherry
Cottonwood
Ash
Cypress
Red Elm
Gum
Hackberry
```

```
White Oak
Hickory
Pecan
Hard Maple
White Oak
Soft Maple
Red Oak
Red Oak
White Oak
Poplar
Sassafras
Sycamore
Black Walnut
Willow
```

2.5 样例输出

```
Ash 13.7931
Aspen 3.4483
Basswood 3.4483
Beech 3.4483
Black Walnut 3.4483
Cherry 3.4483
Cottonwood 3.4483
Cypress 3.4483
Gum 3.4483
Hackberry 3.4483
Hard Maple 3.4483
Hickory 3.4483
Pecan 3.4483
Poplar 3.4483
Red Alder 3.4483
Red Elm 3.4483
Red Oak 6.8966
Sassafras 3.4483
Soft Maple 3.4483
Sycamore 3.4483
White Oak 10.3448
Willow 3.4483
Yellow Birch 3.4483
```

2.6 提示

This problem has huge input, use scanf instead of cin to avoid time limit exceeded.

2.7 来源

Waterloo Local 2002.01.26

3 C: Safecracker

总时间限制: 1000ms; 内存限制: 65536kB

3.1 描述

The item is locked in a Klein safe behind a painting in the second-floor library. Klein safes are extremely rare; most of them, along with Klein and his factory, were destroyed in World War II. Fortunately old Brumbaugh from research knew Klein's secrets and wrote them down before he died. A Klein safe has two distinguishing features: a combination lock that uses letters instead of numbers, and an engraved quotation on the door. A Klein quotation always contains between five and twelve distinct uppercase letters, usually at the beginning of sentences, and mentions one or more numbers. Five of the uppercase letters form the combination that opens the safe. By combining the digits from all the numbers in the appropriate way you get a numeric target. (The details of constructing the target number are classified.) To find the combination you must select five letters v , w , x , y , and z that satisfy the following equation, where each letter is replaced by its ordinal position in the alphabet ($A=1$, $B=2$, ..., $Z=26$). The combination is then $vwxyz$. If there is more than one solution then the combination is the one that is lexicographically greatest, i.e., the one that would appear last in a dictionary.

$$v - w^2 + x^3 - y^4 + z^5 = \text{target}$$

For example, given target 1 and letter set ABCDEFGHIJKL, one possible solution is FIECB, since $6 - 9^2 + 5^3 - 3^4 + 2^5 = 1$. There are actually several solutions in this case, and the combination turns out to be LKEBA. Klein thought it was safe to encode the combination within the engraving, because it could take months of effort to try all the possibilities even if you knew the secret. But of course computers didn't exist then.

Develop a program to find Klein combinations in preparation for field deployment. Use standard test methodology as per departmental regulations.

3.2 输入

Input consists of one or more lines containing a positive integer target less than twelve million, a space, then at least five and at most twelve distinct uppercase letters. The last line will contain a target of zero and the letters END; this signals the end of the input.

3.3 输出

For each line output the unique Klein combination, or 'no solution' if there is no correct combination. Use the exact format shown below.

3.4 样例输入

```
1 ABCDEFGHIJKL
11700519 ZAYEXIWOVU
3072997 SOUGHT
1234567 THEQUICKFROG
0 END
```

3.5 样例输出

```
LKEBA
YOXUZ
GHOST
no solution
```

3.6 来源

Mid-Central USA 2002

4 D: 购买优惠券

总时间限制: 7400ms, 单个测试点时间限制: 200ms; 内存限制: 65536kB

4.1 描述

在超市的货架上有 n 个商品, 每个商品都有一个价值 a_i 。小明有 m 张优惠券, 可以使用这些优惠券购买货架上连续的总价值不超过优惠券面值的若干商品。根据特殊的规定, 小明必须购买 m 张面值相同的优惠券, 而且优惠券面值越高, 价格越贵。请帮助小明计算, 如果他需要购买所有商品, 最少需要购买的优惠券面值是多少。

优惠券面值必须是整数, 且一个商品不能用两张优惠券凑起来购买。比如一张优惠券购买了若干商品后还剩 2 元, 然而下一个商品价格超过 2 元, 那么这张优惠券剩下的 2 元只能作废。

请注意: 所有商品价格之和可能超过 `int` 表示范围, 所以答案也可能超过 `int` 表示范围。该用 `long long` 的地方就要用 `long long`。

4.2 输入

第一行两个整数 n, m , 分别表示商品的数量 ($1 \leq n \leq 10^5$) 和要分成的组数 ($1 \leq m \leq n$)。第二行 n 个整数 $a_1, a_2, \dots, a_n$, 表示每个商品的价值 ($1 \leq a_i \leq 10^5$)

4.3 输出

一个整数, 表示最少需要购买的优惠券面值

4.4 样例输入

```
7 3
2 5 1 9 3 6 2
```

4.5 样例输出

```
11
```

4.6 提示

分别购买 $[2, 5, 1]$, $[9]$, $[3, 6, 2]$

5 E: 合并编码

总时间限制: 13000ms, 单个测试点时间限制: 1000ms; 内存限制: 262144kB

5.1 描述

给定一个合并编码后的字符串, 返回原始字符串。

编码规则为: $k[\text{string}]$, 表示其中方括号内部的 `string` 重复 k 次, k 为正整数且 $0 < k < 20$ 。可以认为输入数据中所有的数字只表示重复的次数 k , 且方括号总是满足要求的, 例如不会出现像 `3a`、`2[b` 或 `2[4]` 的输入。

输入保证符合格式, 且不包含空格。

5.2 输入

一行, 一个长度为 n ($1 \leq n \leq 500$) 的字符串, 代表合并编码后的字符串。

5.3 输出

一行, 代表原始字符串。长度 m 满足 $0 \leq m \leq 1500$ 。

5.4 样例输入 1

```
3[abc]1[o]2[n]
```

5.5 样例输出 1

```
abcabcabconn
```

5.6 样例输入 2

```
3[a2[c]]
```

5.7 样例输出 2

```
accaccacc
```


6 F: 网线主管

总时间限制: 1000ms; 内存限制: 65536kB

6.1 描述

仙境的居民们决定举办一场程序设计区域赛。裁判委员会完全由自愿组成，他们承诺要组织一次史上最公正的比赛。他们决定将选手的电脑用星形拓扑结构连接在一起，即将它们全部连到一个单一的中心服务器。为了组织这个完全公正的比赛，裁判委员会主席提出要将所有选手的电脑等距离地围绕在服务器周围放置。

为购买网线，裁判委员会联系了当地的一个网络解决方案提供商，要求能够提供一定数量的等长网线。裁判委员会希望网线越长越好，这样选手们之间的距离可以尽可能远一些。

该公司的网线主管承接了这个任务。他知道库存中每条网线的长度（精确到厘米），并且只要告诉他所需的网线长度（精确到厘米），他都能够完成对网线的切割工作。但是，这次，所需的网线长度并不知道，这让网线主管不知所措。

你需要编写一个程序，帮助网线主管确定一个最长的网线长度，并且按此长度对库存中的网线进行切割，能够得到指定数量的网线。

6.2 输入

第一行包含两个整数 N 和 K ，以单个空格隔开。 N ($1 \leq N \leq 10000$) 是库存中的网线数， K ($1 \leq K \leq 10000$) 是需要的网线数量。接下来 N 行，每行一个数，为库存中每条网线的长度（单位：米）。所有网线的长度至少 1m，至多 100km。输入中的所有长度都精确到厘米，即保留到小数点后两位。

6.3 输出

网线主管能够从库存的网线中切出指定数量的网线的最长长度（单位：米）。必须精确到厘米，即保留到小数点后两位。若无法得到长度至少为 1cm 的指定数量的网线，则必须输出 "0.00"（不包含引号）。

6.4 样例输入

```
4 11
8.02
7.43
4.57
5.39
```

6.5 样例输出

```
2.00
```

6.6 来源

Northeastern Europe 2001

7 G: Dungeon Master

总时间限制: 1000ms; 内存限制: 65536kB

7.1 描述

You are trapped in a 3D dungeon and need to find the quickest way out! The dungeon is composed of unit cubes which may or may not be filled with rock. It takes one minute to move one unit north, south, east, west, up or down. You cannot move diagonally and the maze is surrounded by solid rock on all sides.

Is an escape possible? If yes, how long will it take?

7.2 输入

The input consists of a number of dungeons. Each dungeon description starts with a line containing three integers L, R and C (all limited to 30 in size). L is the number of levels making up the dungeon. R and C are the number of rows and columns making up the plan of each level. Then there will follow L blocks of R lines each containing C characters. Each character describes one cell of the dungeon. A cell full of rock is indicated by a # and empty cells are represented by a . . Your starting position is indicated by S and the exit by the letter E. There's a single blank line after each level. Input is terminated by three zeroes for L, R and C.

7.3 输出

Each maze generates one line of output. If it is possible to reach the exit, print a line of the form **Escaped in x minute(s)** ., where x is replaced by the shortest time it takes to escape.

If it is not possible to escape, print the line **Trapped!** .

7.4 样例输入

```
3 4 5
S...
.###.
.##..
###.#

#####
#####
##.##
##...

#####
#####
#.###
####E

1 3 3
```

```
S##  
#E#  
###  
  
0 0 0
```

7.5 样例输出

```
Escaped in 11 minute(s).  
Trapped!
```

7.6 来源

Ulm Local 1997

8 H: 糖果

总时间限制: 1000ms; 内存限制: 65536kB

8.1 描述

由于在维护世界和平的事务中做出巨大贡献，Dzx 被赠予糖果公司 2010 年 5 月 23 日当天无限量糖果免费优惠券。在这一天，Dzx 可以从糖果公司的 N 件产品中任意选择若干件带回家享用。糖果公司的 N 件产品每件都包含数量不同的糖果。Dzx 希望他选择的产品包含的糖果总数是 K 的整数倍，这样他才能平均地将糖果分给帮助他维护世界和平的伙伴们。当然，在满足这一条件的基础上，糖果总数越多越好。Dzx 最多能带走多少糖果呢？注意：Dzx 只能将糖果公司的产品整件带走。

8.2 输入

第一行包含两个整数 $N(1 \leq N \leq 100)$ 和 $K(1 \leq K \leq 100)$ 以下 N 行每行 1 个整数，表示糖果公司该件产品中包含的糖果数目，不超过 1000000

8.3 输出

符合要求的最多能达到的糖果总数，如果不能达到 K 的倍数这一要求，输出 0

8.4 样例输入

```
5 7
1
2
3
4
5
```

8.5 样例输出

```
14
```

8.6 提示

Dzx 的选择是 $2+3+4+5=14$ ，这样糖果总数是 7 的倍数，并且是总数最多的选择。